Research Article | Vol. 17, No. 9 / 1 Sep 2026 / *Biomedical Optics Express* 4933

Biomedical Optics EXPRESS

DASHING–LSCI: deep learning-based anatomical multiclass segmentation for human intraoperative neurosurgical guidance – laser speckle contrast imaging

Ebenezer Raj Selvaraj Mercyshalinie,[1] Paul Calle,[1] Jacqueline O. Stovall,[1] Brayden J. Girard,[1] Diogo Pinheiro Cordeiro,[1] Andrew M. Bauer,[2] Johannes Goldberg,[3] Christopher S. Graffeo,[2] Andreas Raabe,[3] Michael T. Lawton,[4] Andrew K. Dunn,[5] Chongle Pan,[1,6] and David R. Miller[1,2,*]

[1] *Stephenson School of Biomedical Engineering, University of Oklahoma, Norman, Oklahoma, USA*
[2] *Department of Neurosurgery, University of Oklahoma Health Sciences Center, Oklahoma City, Oklahoma, USA*
[3] *Department of Neurosurgery, Inselspital, Bern University Hospital, University of Bern, Freiburgstrasse, Bern, Switzerland*
[4] *Department of Neurosurgery, Barrow Neurological Institute, St. Joseph's Hospital and Medical Center, Phoenix, Arizona, USA*
[5] *Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX, USA*
[6] *School of Computer Science, University of Oklahoma, Norman, Oklahoma, USA*
[*] *dmiller@ou.edu*

Abstract: Laser speckle contrast imaging (LSCI) provides real-time, label-free visualization of cerebral blood flow in vessels and perfusion in cortical tissue during neurosurgery. At present, surgeons interpret LSCI-derived flow and perfusion changes qualitatively between microsurgical maneuvers, which limits continuous quantitative analysis. Automated delineation of blood vessels and cortical tissue is a prerequisite for such analysis. Here we evaluate a deep learning approach for multiclass segmentation of intraoperative LSCI images into blood vessels, cortical tissue, and background (e.g., surgical instruments, skull). We curated 90 intraoperative LSCI images acquired during 18 unique human neurosurgery procedures across three prospective observational clinical studies (30 images per site) and trained nnU-Net (Version 2) to segment images. To assess cross-site generalization, one site was reserved exclusively for testing while the images from the other two were pooled and split into training and validation sets. Across three held-out sites, the model achieved mean Dice scores of 0.88 for background and 0.73 for cortical tissue, but a lower and more variable 0.41 for blood vessels, reflecting the difficulty of segmenting thin, sparse vessel structures. These results demonstrate the feasibility of deep-learning segmentation for extracting anatomical structure from neurosurgical LSCI images, a capability needed for real-time, continuous blood flow and perfusion analysis to support surgical guidance.

1. Introduction

Neurosurgical interventions, particularly those involving microsurgery, require high precision and accuracy to navigate intricate brain structures while preserving vital tissues [1,2]. To better characterize real-time cerebrovascular dynamics, laser speckle contrast imaging (LSCI) has emerged as a tool for visualizing blood flow and tissue perfusion continuously and label-free during neurosurgical procedures. We [3–8] and others [9–12] have demonstrated the potential

#605293 https://doi.org/10.1364/BOE.605293
Journal © 2026 Received 18 May 2026; revised 23 Jul 2026; accepted 19 Aug 2026; published 1 Sep 2026

Biomedical Optics EXPRESS

for LSCI to provide cerebral blood flow visualization during neurosurgery; however, these studies analyzed the LSCI-derived blood flow data after surgery rather than in real time. While LSCI provides visualization of real-time and continuous full-field blood flow maps, frequent manual interpretation of these maps by the operating surgeon while performing microsurgery is not realistic. Instead, an automated analysis of LSCI images is needed which can alert the operating surgeon when visual inspection is needed, such as to highlight potential ischemic risks or hyperperfusion. We aim to develop an automated, real-time method to segment blood vessels and cortical tissue for LSCI images during neurosurgery, which we term DASHING–LSCI (Deep learning-based Anatomical multiclass Segmentation for Human Intraoperative Neurosurgical Guidance – Laser Speckle Contrast Imaging).

LSCI enables assessment of vessel patency, reperfusion, and tissue viability during neurosurgery procedures such as cerebral aneurysm clipping [8], cerebral bypass [10,11], arteriovenous malformation (AVM) resection [6], and brain tumor resection [8,12]. Extracting reliable flow information from LSCI requires segmenting blood vessels and cortical gyrus regions. Automated vessel segmentation has been studied extensively in other modalities, including CT and MR angiography, digital subtraction angiography, ultrasound, retinal fundus photography, and optical coherence tomography [13–19]. These methods are tuned to image-formation and noise characteristics that differ fundamentally from intraoperative LSCI, and most treat segmentation as a binary vessel-versus-background problem. Deep-learning work specific to LSCI has likewise focused on binary cerebral vessel segmentation, often using unsupervised domain adaptation without ground truth labels [20] or incorporating experimental setup information to improve generalization [21]. However, effective surgical guidance requires more than binary vessel detection: separating blood vessels from surrounding cortical tissue and from non-anatomical background is necessary for region-specific perfusion analysis with LSCI. Multiclass segmentation of intraoperative LSCI is further complicated by pulsatile tissue motion and specular reflections [8,22].

To address these challenges, we adopt nnU-Net (Version 2), a self-configuring method that automatically tailors preprocessing, network architecture, and training parameters to dataset specifics, lowering the technical barriers to clinical use [23,24]. While previous LSCI approaches have largely focused on binary segmentation, we trained and evaluated nnU-Net for multiclass segmentation of intraoperative LSCI images into three classes: blood vessels, cortical tissue, and background. This proposed methodology is designed to provide neurosurgeons with real-time region-specific blood flow and tissue perfusion analysis in an effort to provide an automated, real-time system that can alert surgeons to developing ischemia, hyperperfusion, or perfusion changes after anastomosis.

2. Materials and methods

2.1. Data collection

To ensure the deep learning model was robust and generalizable, we utilized intraoperative LSCI recordings acquired at three prospective observational clinical study sites: Site 1 is the Barrow Neurological Institute in Phoenix, AZ, USA; Site 2 is Inselspital Bern University Hospital in Bern, Switzerland; and Site 3 is the University of Oklahoma Health Sciences Center in Oklahoma City, OK, USA. The Site 1 study was approved by the Dignity Health Institutional Review Board (ClinicalTrials.gov identifier: NCT05305378); the Site 2 study was approved by the local ethics committee in Bern, Switzerland (Project-ID: 2021-D0043), and the Site 3 study was approved by the University of Oklahoma Health Sciences Center's Institutional Review Board (IRB# 18049). All studies were conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [25]. Recordings were obtained during cerebral aneurysm clipping, cerebral bypass, and brain tumor resection procedures between 2022 and 2025 from a total of 18 unique patients. An LSCI frame was included if it met two pre-defined

Biomedical Optics EXPRESS

criteria: 1) it contained a full view of the exposed cortical surface, corresponding to a microscope magnification of 2–4×, and 2) no surgical instruments obstructed the field of view of the cortex. LSCI frames were not otherwise selected based on image quality, segmentation difficulty, or model performance. From each site, we included the first 30 qualifying frames, resulting in 90 images total; the number of images used per patient is shown in Table 1. Site-level separation was maintained across the training, validation, and test splits; since no patient was imaged at more than one site, holding out an entire site for testing ensures that no test-set patient images appear in the training or validation data. Within the training and validation set of images, the split was performed at the image level, so images from the same patient could appear in both the training and validation subsets.

Table 1. Patient-level composition of the dataset and leave-one-site-out split assignments

Patient	Site	Images	Split 1	Split 2	Split 3
BNI-1	1	1	**Test**	Train/val	Train/val
BNI-2	1	11	**Test**	Train/val	Train/val
BNI-3	1	2	**Test**	Train/val	Train/val
BNI-4	1	6	**Test**	Train/val	Train/val
BNI-5	1	10	**Test**	Train/val	Train/val
Insel-1	2	5	Train/val	**Test**	Train/val
Insel-2	2	1	Train/val	**Test**	Train/val
Insel-3	2	2	Train/val	**Test**	Train/val
Insel-4	2	5	Train/val	**Test**	Train/val
Insel-5	2	1	Train/val	**Test**	Train/val
Insel-6	2	9	Train/val	**Test**	Train/val
Insel-7	2	5	Train/val	**Test**	Train/val
Insel-8	2	1	Train/val	**Test**	Train/val
Insel-9	2	1	Train/val	**Test**	Train/val
OU-1	3	8	Train/val	Train/val	**Test**
OU-2	3	7	Train/val	Train/val	**Test**
OU-3	3	9	Train/val	Train/val	**Test**
OU-4	3	6	Train/val	Train/val	**Test**

2.2. Instrumentation

The data preparation process is visualized in Fig. 1. We display the internal microscope camera image of the surgical field (Fig. 1(A)) to show the surgeon's view and the co-registered LSCI image (Fig. 1(B)). Sites 1 and 3 used the neurosurgical microscope Kinevo 900 (Carl Zeiss Meditec, Oberkochen, Germany) and Site 2 used a Pentero 900 (Carl Zeiss Meditec, Oberkochen, Germany). All sites used a custom microscope-integrated LSCI device mounted onto the surgical microscope without interfering with its normal use, as detailed in previous publications [7] and [8]. A λ=785 nm laser diode (L785P090, Thorlabs, Newton, NJ) with maximum output power of 100 mW was secured to a custom adapter mounted on the underside of the microscope. The emitted light reflected off a mirror, diverging to an approximate 4 cm beam size at a 35 cm working distance. The maximum irradiance delivered to tissue was 0.01 W/cm^2, well below the American National Standards Institute limit of 0.3 W/cm^2 for skin exposure at 785 nm [26]. The microscope's imaging optics collected the back-scattered laser light, directing it via a dichroic mirror to an NIR-enhanced camera (acA1300-60gmNIR, Basler AG, Ahrensburg, Germany),

Biomedical Optics EXPRESS

operating at a frame rate of 60 fps. A laser controller (LDC205C, Thorlabs, Newton, New Jersey, USA) ensured the laser remained on exclusively during LSCI recordings. The microscope parameters, including magnification and the subsequent imaging field of view, were dictated by the surgeon. For the exposed cortex images presented in this paper, magnification typically ranged from 2–4× with a 3–5 cm field of view. The speckle-to-pixel ratio is ~4 at a magnification of 3×, satisfying the Nyquist sampling criterion (which requires a ratio of at least 2) and avoiding underestimation of the speckle contrast [27]. To assess tissue perfusion in real time during surgery, the microscope-integrated camera captures raw speckle images and converts them into LSCI images by calculating the spatial speckle contrast, K, within a 7×7 pixel moving window using the equation $K = \sigma_s/I$, where σ_s represents the spatial standard deviation and I is the average intensity within that region. We did not apply any other post-processing techniques following the calculation of K. We used a single-exposure time of 3 ms, selected based on prior studies demonstrating that 0.5, 3, and 5 ms exposure times most closely match the results of multiple exposure times over the range of 0.5 to 20 ms [28].

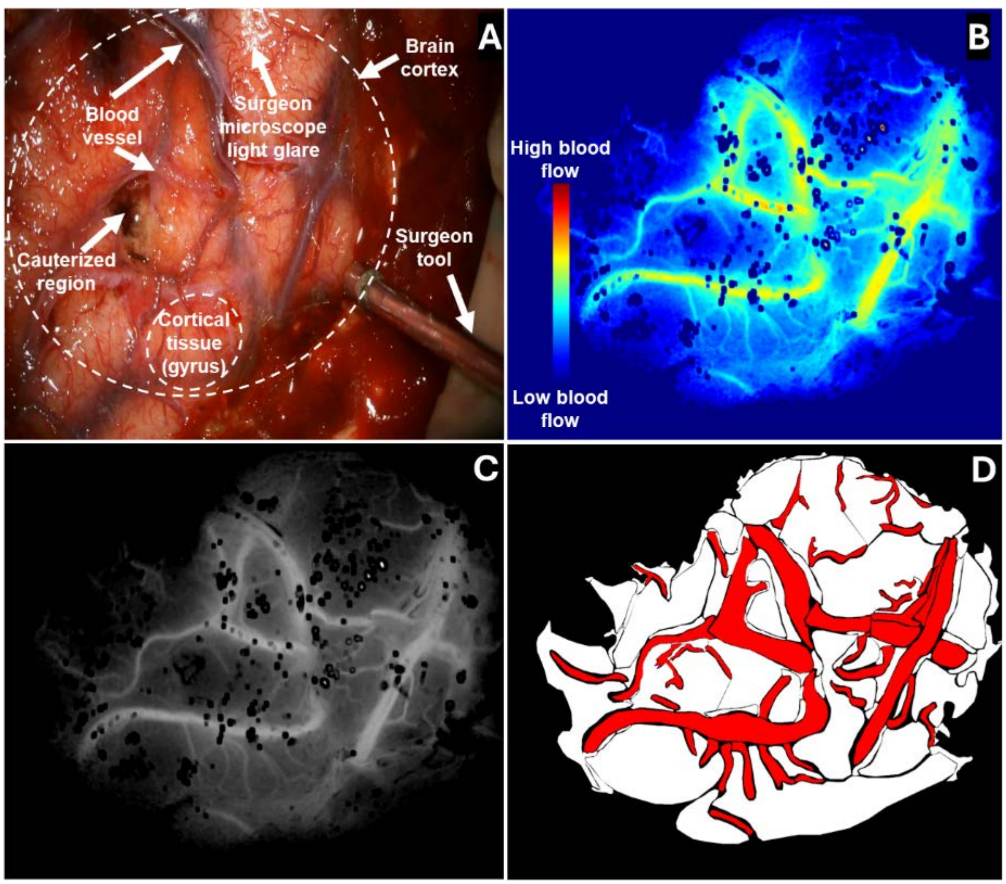

Fig. 1. Illustrative figure with example images and ground truth labels. (A) White light image from the internal camera of the surgical microscope showing the surgical field of view. (B) Laser speckle contrast image (LSCI) co-registered with the microscope internal camera shown with a jet colormap. (C) Grayscale LSCI image shown for visualization purposes. (D) Human-labeled ground truth image where red indicates blood vessels, white indicates cortical tissue, and black indicates background.

Biomedical Optics EXPRESS

2.3. Image preprocessing and labeling

We generated LSCI frames from the recorded intraoperative speckle contrast data and manually labeled anatomical structures. A single LSCI frame was produced for labeling by averaging multiple speckle contrast images over a 0.5–1 second temporal window. For input to the deep learning model, each frame was rendered with a grayscale colormap over a speckle contrast range of 0.00–0.20; no resizing or down-sampling was applied. For visualization, Fig. 1 shows example frames rendered with a jet colormap (Fig. 1(B)) and a grayscale colormap (Fig. 1(C)). Ground truth labels were produced by a single annotator using LabelMe [29] in Python to draw the polygonal regions defining each class (Fig. 1(D)). Based on physiological structure, surgical relevance, and input from neurosurgeons, the segmentation masks comprised three classes: blood vessels (red in Fig. 1(D)), cortical tissue (white in Fig. 1(D)), and background which encompasses the skull, surgeon's hands, and surgical tools (black in Fig. 1(D)). Annotations were exported as JSON files containing the polygon labels and then converted to PNG segmentation masks.

2.4. Implementation of nnU-Net

We conducted training of the nnU-Net (Version 2) model on a high-performance system running Ubuntu 20.04.6 LTS, equipped with two NVIDIA RTX A6000 GPUs (48 GB memory each) utilizing CUDA 12.4. Model evaluation employed a three-fold site-level hold-out strategy. In each iteration, data from one institution (30 images) was reserved exclusively for testing, while data from the remaining two institutions (60 images) were pooled. This pooled data was randomly split at the image level into 48 images for training and 12 images for validation (an 80/20 split). The validation set was used exclusively to monitor training progress and identify the optimal model checkpoint based on the exponential moving average of the validation Dice score. The network architecture and training hyperparameters were fixed in a static 2D configuration established prior to training; no dynamic hyperparameter tuning was performed during the loops. The batch size was fixed at 2 with a patch size of [1024,1536]. The architecture was a PlainConvUNet with 9 stages, with feature maps starting at 32 and scaling up to 512. In all experiments, the model was trained for 200 epochs. Data augmentation was performed automatically by nnU-Net's built-in data augmentation pipeline. For all runs, we utilized the default hyperparameter configuration of the nnU-Net Version 2 framework to ensure reproducibility and leverage the model's self-configuring capabilities. The training process employed an initial learning rate of 1×10^{-2} and a weight decay of 3×10^{-5}. To address potential class imbalance, a foreground oversampling rate of 0.33 was applied. Deep supervision was enabled by default to facilitate gradient propagation during training. Each training epoch consisted of 250 iterations, with 50 iterations dedicated to validation. These parameters were kept consistent across all site data splits.

2.5. Evaluation metrics

We evaluated segmentation performance using two complementary groups of metrics. Overlap and pixel-classification agreement between predicted masks and ground-truth labels were quantified with the Dice similarity coefficient, intersection over union (IoU), precision, recall, and specificity [23,30]. The Dice similarity coefficient measures regional overlap and equals twice the number of correctly predicted class pixels divided by the total number of class pixels in the predicted and ground-truth masks combined. IoU is the ratio of the intersection to the union of the predicted and ground-truth regions and penalizes errors more strictly than Dice. Precision is the fraction of pixels predicted as a given class that truly belong to it, so it decreases with over-segmentation (false positives), whereas recall, or sensitivity, is the fraction of true class pixels that are correctly identified and decreases with under-segmentation (missed pixels). Specificity is the fraction of non-class pixels correctly excluded. Because these overlap-based metrics can be misleading for thin, sparse structures such as blood vessels, we additionally assessed how well the segmentations preserved vessel topology using the centerline Dice [31] and the Betti-0 and Betti-1 errors, which

Biomedical Optics EXPRESS

measure discrepancies in the number of connected components and loops, respectively, relative to the ground truth [32]. Topological precision and topological sensitivity denote the fractions of the predicted vessel skeleton lying within the ground-truth vessel mask and of the ground-truth skeleton lying within the predicted mask, respectively, and the centerline Dice is their harmonic mean. All metrics were computed per image and reported per class.

2.6. Statistical analysis

Between-class differences in per-image performance were evaluated for all five metrics (Dice, IoU, precision, recall, and specificity) using paired t-tests, which are appropriate here because each test image contributes one value per class ($n = 30$ paired images per site). To account for the three pairwise class comparisons within each metric at each site, significance was assessed against a Bonferroni-corrected threshold of $\alpha = 0.05/3 = 0.0167$. Because Dice, IoU, and the other bounded metrics can have non-normally distributed differences, every comparison was additionally repeated with a Wilcoxon signed-rank test.

3. Results

3.1. Training and validation statistics

Figure 2 illustrates the training dynamics. In each split, data from two institutions were pooled and randomly partitioned at the image level into training and validation sets, while the remaining institution was held out for testing. In the first split (Fig. 2(A) and Fig. 2(B)), pooled data from Sites 2 and 3 were used for training and validation. The loss curves demonstrate convergence, and the mean validation Dice score stabilized as training progressed. In the second split (Fig. 2(C) and Fig. 2(D)), the model was trained and validated on pooled data from Sites 1 and 3, exhibiting similar convergence behavior and progressive improvement in validation Dice. In the third split (Fig. 2(E) and Fig. 2(F)), pooled data from Sites 1 and 2 were used for training and validation, again demonstrating consistent loss reduction and stabilization of validation performance. Across all three scenarios, the training and validation loss decreased consistently, and the validation Dice scores improved, indicating successful model convergence without significant overfitting.

3.2. Quantitative performance and generalization on testing data

Class-wise segmentation performance on the held-out test sets from the three sites is summarized in Fig. 3. For the background class, the mean Dice score was 0.88, with per-site Dice scores of 0.90 (Site 1), 0.95 (Site 2), and 0.78 (Site 3), with corresponding IoU of 0.82, 0.91, and 0.66; precision of 0.97, 0.98, and 0.77; recall of 0.84, 0.93, and 0.80; and specificity of 0.92, 0.91, and 0.85. The high background scores reflect both the visual distinctness of this class and its prevalence, as 69.4% of all pixels across the 90 images belong to the background. For cortical tissue, the mean Dice score was 0.73, with per-site Dice scores of 0.66, 0.73, and 0.79 across Sites 1–3, with IoU of 0.50, 0.59, and 0.66; precision of 0.55, 0.65, and 0.80; recall of 0.87, 0.88, and 0.80; and specificity of 0.84, 0.90, and 0.80. Blood vessels were the most challenging class, with a mean Dice score of 0.41 and per-site Dice scores of 0.39, 0.36, and 0.49; IoU of 0.25, 0.23, and 0.33; precision of 0.41, 0.66, and 0.60; recall of 0.48, 0.27, and 0.45; and specificity of 0.98, 0.99, and 0.97. The consistently high vessel specificity does not indicate strong vessel detection; rather, it follows from the small proportion of vessel pixels, which makes true negatives dominate. For this thin, sparse class, the lower Dice, IoU, and recall are the more informative measures.

Between-class differences in per-image performance were assessed for all five metrics; the paired t-test and Wilcoxon signed-rank test yielded the same significance conclusions in every comparison. Complete per-metric p-values are provided in the Supplement 1 (Tables S1–S3). At Sites 1 and 2, the spatial-overlap metrics (Dice and IoU) differed significantly across all three class pairs, confirming that background, cortical tissue, and blood vessels are segmented at

Biomedical Optics EXPRESS

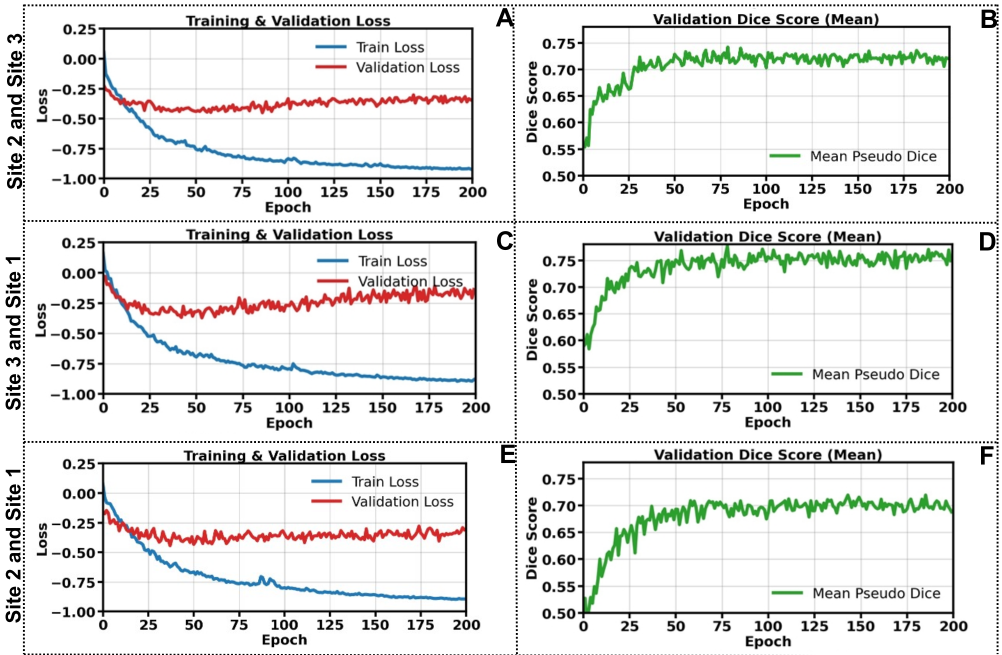

Fig. 2. Training and validation performance across site-level hold-out splits. Training and validation metrics over 200 epochs for three site-level splits. Training and validation loss (A, C, E) and mean validation Dice score (B, D, F) for data pooled from Sites 2 and 3, Sites 1 and 3, and Sites 1 and 2, respectively.

distinct levels of overlap accuracy. The non-significant differences that did occur were limited to precision, recall, and specificity and reflect comparable performance between the classes involved rather than a segmentation failure: at Site 1, recall did not differ between cortical tissue and background ($p = 0.10$), and at Site 2, precision did not differ between blood vessels and cortical tissue ($p = 0.82$), nor did recall or specificity differ between cortical tissue and background ($p = 0.13$ and $p = 0.68$, respectively). At Site 3, blood vessels differed significantly from both cortical tissue and background on all five metrics, whereas cortical tissue and background did not differ on any metric (all $p \geq 0.066$), consistent with these two non-vessel classes being segmented with similarly high accuracy at that site.

Figure 4 reports blood vessel specific and topology-aware metrics for the blood-vessel class. Mean centerline Dice was 0.45 (Site 1), 0.46 (Site 2), and 0.54 (Site 3). Mean topological precision was 0.45, 0.74, and 0.64, and mean topological sensitivity was 0.54, 0.35, and 0.51, respectively. Mean Betti-0 error was 22.73 (median 25.0) at Site 1, 4.87 (median 3.5) at Site 2, and 9.73 (median 10.0) at Site 3. Mean Betti-1 error was 4.07 (median 3.5), 2.30 (median 1.0), and 5.67 (median 4.0), respectively. Whereas centerline overlap was relatively consistent across sites (0.45–0.54), topological fidelity was not. Site 2 produced the lowest topological errors and Site 1 the highest Betti-0 error. At Site 2, vessels were predicted with high precision but low sensitivity, giving few false branches but many missed vessels, consistent with its low Betti errors and the fewest topological inconsistencies. Site 1 showed the opposite pattern, with low precision and high sensitivity, producing more fragmented or spurious components and the largest Betti-0 error. Site 3 obtained the highest centerline overlap yet also the largest Betti-1 error, indicating that strong pixel-level agreement does not guarantee a topologically correct segmentation. Overall, topological fidelity rather than pixel-level overlap remains the primary limitation for the vessel class, and this limitation was strongly dependent on site.

Biomedical Optics EXPRESS

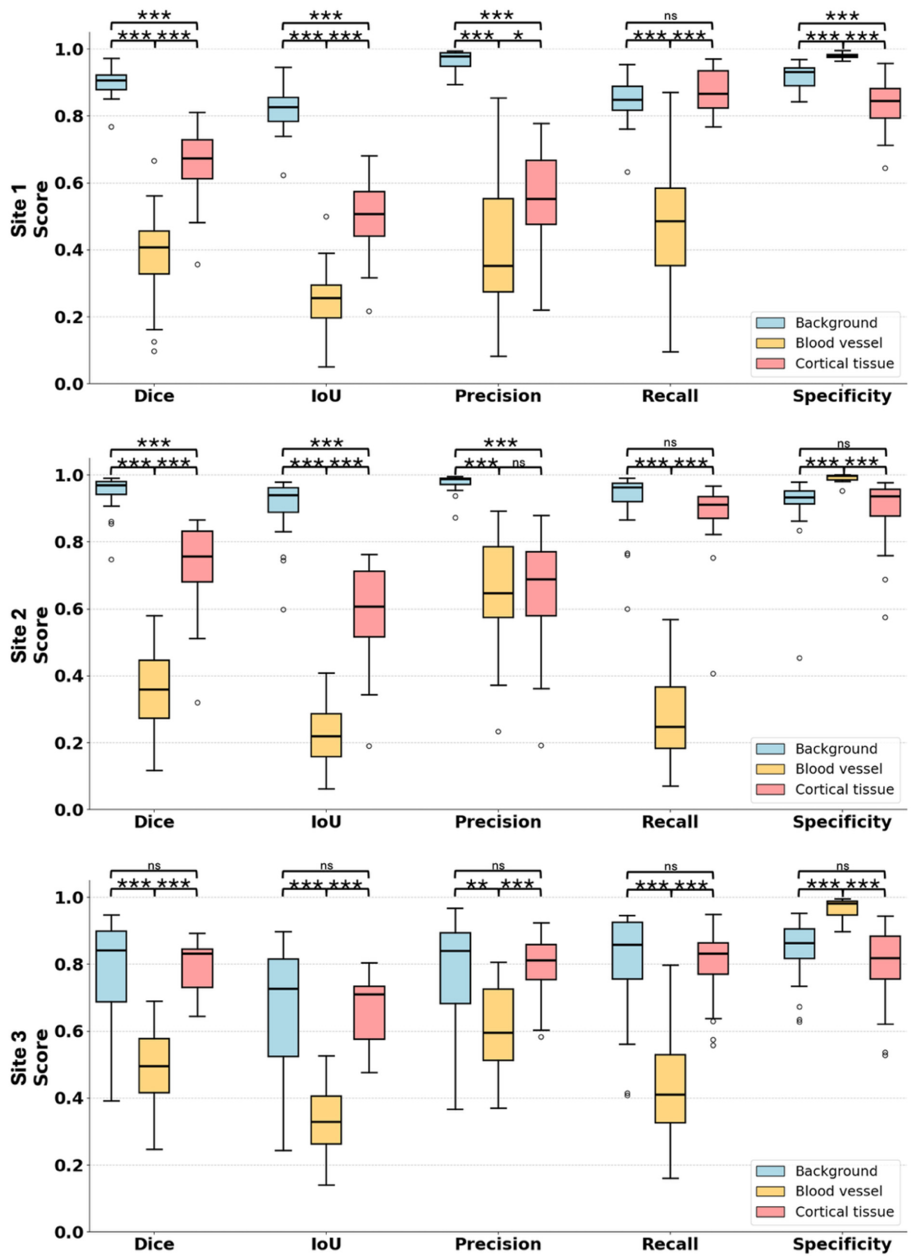

Fig. 3. Class-wise segmentation performance on the held-out test data across the three clinical sites. Each panel shows the Dice similarity coefficient, intersection over union (IoU), precision, recall, and specificity for the three segmentation classes: background (blue), blood vessels (yellow), and cortical tissue (red), across Site 1 (top), Site 2 (middle), and Site 3 (bottom). In each box plot, the central line denotes the median, the box spans the interquartile range (25^{th}–75^{th} percentiles), whiskers extend to the most extreme values within 1.5× the interquartile range, and individual points denote outliers ($n = 30$ images per site). Brackets indicate pairwise between-class comparisons (paired t-test with Bonferroni correction, $\alpha = 0.0167$): ns, not significant; *$p < 0.0167$; **$p < 10^{-3}$; ***$p < 10^{-4}$. A Wilcoxon signed-rank test yielded the same significance conclusions; complete per-metric p-values are provided in the Supplement 1.

Biomedical Optics EXPRESS

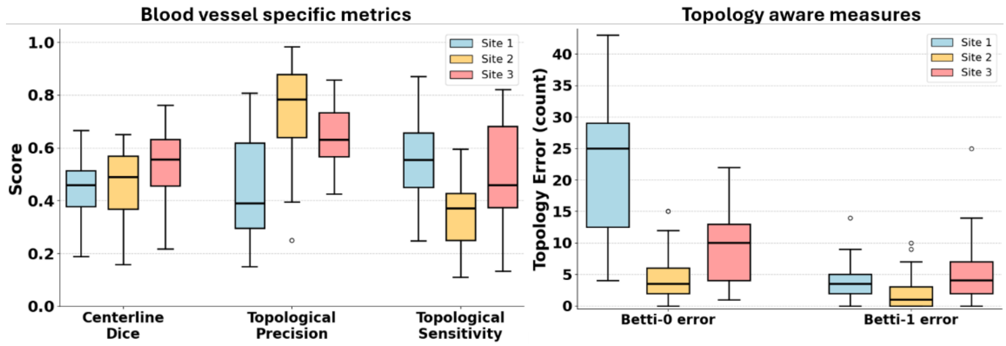

Fig. 4. Blood-vessel-specific evaluation and topology-aware segmentation performance for the blood vessel class across the three clinical sites. **Left**: centerline Dice, topological precision, and topological sensitivity, which quantify how well the predicted vessel skeleton overlaps and reproduces the reference vessel topology. **Right**: topological errors, reported as the Betti-0 error (mismatch in the number of connected components, reflecting vessel fragmentation) and Betti-1 error (mismatch in the number of loops). In each box plot, the central line is the median, the box spans the interquartile range (25^{th}–75^{th} percentiles), whiskers extend to the most extreme values within 1.5× the interquartile range, and points denote outliers; colors indicate Site 1 (blue), Site 2 (yellow), and Site 3 (red) ($n = 30$ images per site).

3.3. Qualitative assessment of nnU-Net segmentation across multi-center datasets

To qualitatively assess the clinical utility of the model, we compared the nnU-Net predictions against the original LSCI images and the human-labeled ground truth across the three testing sites (Fig. 5). For the Site 1 data (Fig. 5(A)), three representative examples (Fig. 5(A1–A3), Fig. 5(A4–A6), Fig. 5(A7–A9) show that the predicted labels (Fig. 5A3, 5A6, 5A9) match the human-labeled ground truth (Fig. 5A2, 5A5, 5A8) and delineate the vasculature visible in the LSCI images (Fig. 5A1, 5A4, 5A7). One failure case is shown in Fig. 5(A9), where the model incorrectly labeled the top-right corner as cortical tissue; this was caused by a small cluster of saturated pixels in that region. Saturation arises occasionally during acquisition, depends on instrumentation and imaging conditions, and was not specific to any one site. For the Site 2 data (Fig. 5(B)), the LSCI images (Fig. 5B1, 5B4, 5B7), ground truth (Fig. 5B2, 5B5, 5B8), and model predictions (Fig. 5B3, 5B6, 5B9) show close agreement, indicating that the model generalized to the image characteristics specific to this site. For the Site 3 data (Fig. 5(C)), the predicted labels (Fig. 5C3, 5C6, 5C9) exhibit high fidelity to the ground truth (Fig. 5C2, 5C5, 5C8) derived from the LSCI images (Fig. 5C1, 5C4, 5C7).

Figure 6 provides a qualitative assessment of the nnU-Net model's segmentation of blood vessels and cortical tissue on the testing data, across the three distinct clinical sites (1, 2 and 3). One image from each site is shown (Fig. 6(A–O)). Across all sites, nnU-Net model predictions align closely with ground truth annotations, exhibiting strong overall agreement across the entire surgical field of view. The model accurately traces most blood vessels and reliably isolates cortical tissue with minimal false positives. These overlapping masks confirm the model's adaptability and high spatial accuracy regardless of site-specific image acquisition variations.

3.4. Computational efficiency

To assess the computational feasibility of intraoperative deployment, we measured the wall-clock inference time required by the trained model to generate the output masks for the held-out test sets. Generating the 30 test masks took 25.18, 13.25, and 15.31 seconds for Sites 1, 2, and

Biomedical Optics EXPRESS

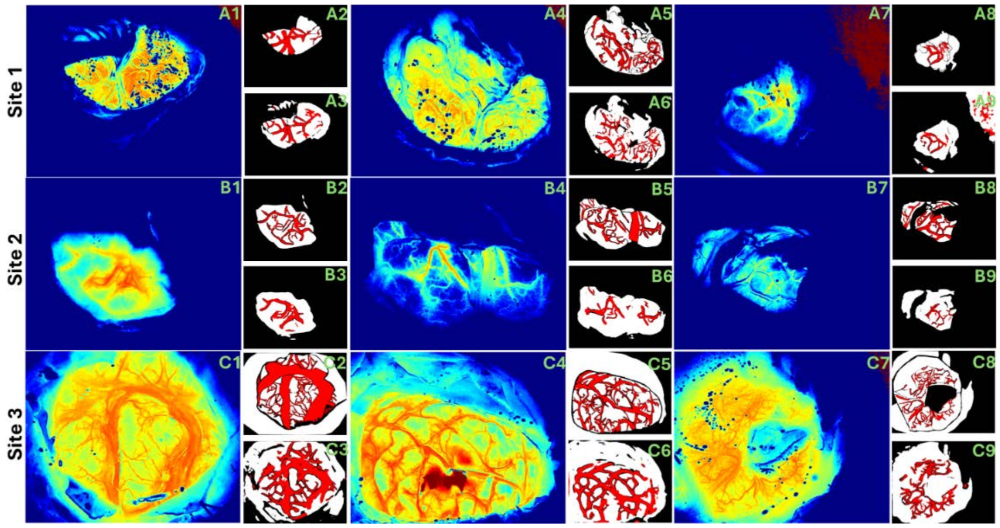

Fig. 5. Visual segmentation results of testing data. Comparison of laser speckle contrast imaging (LSCI), ground truth, and nnU-Net predictions. (A) Site 1 test data: examples 1, 2, and 3 show the LSCI (A1, A4, A7), human-labeled ground truth (A2, A5, A8), and nnU-Net predicted labels (A3, A6, A9). (B) Site 2 test data: examples 1, 2, and 3 show the LSCI (B1, B4, B7), human-labeled ground truth (B2, B5, B8), and nnU-Net predicted labels (B3, B6, B9). (C) Site 3 test data: examples 1, 2, and 3 show the LSCI (C1, C4, C7), human-labeled ground truth (C2, C5, C8), and nnU-Net predicted labels (C3, C6, C9).

3, respectively, corresponding to mean single-image inference times of 0.84, 0.44, and 0.51 s per image (an overall average of $\approx$ 0.60 s per image). At the 60 fps frame rate used in this study, averaging 30 frames produces a new perfusion map every 0.5 seconds, an interval that could be shortened with a sliding-window average. Converting each raw speckle image to a laser speckle contrast image takes approximately 0.33 milliseconds on the 20-core CPU used in this study, following the implementation of the processing algorithms described in our previous publication [33]. Overlaying the inference results on the perfusion map and displaying them on a monitor takes approximately 0.1 seconds. Thus, we estimate that the process to record a 30-frame-averaged perfusion map, perform inferencing, and display the inference results overlaid on the perfusion map on a monitor for the surgeon to observe would take approximately 1.2 seconds. This time amount is clinically feasible for near real-time guidance to the surgeon during intraoperative neurosurgery.

4. Discussion and conclusion

To our knowledge, this study represents the first implementation of the nnU-Net architecture for multiclass segmentation of intraoperative LSCI images. While LSCI provides high-contrast visualization of blood flow, it is characterized by unique noise textures and signal fluctuations that differ significantly from standard MRI or CT modalities. Our primary objective was to validate that a deep learning model can effectively differentiate between blood vessels, cortical tissue, and background elements such as surgical instruments or the skull, without extensive manual preprocessing. Qualitative visual inspection (Figs. 5–6) demonstrates that the model successfully segments complex vessel structures and tissue boundaries, accurately defining the vasculature visible in LSCI images. This is further supported by the model's stability across variations in surgical lighting, microscope configurations, and patient anatomy. The overall mean validation Dice score of 0.70 indicates stable convergence during training. Performance

Biomedical Optics EXPRESS

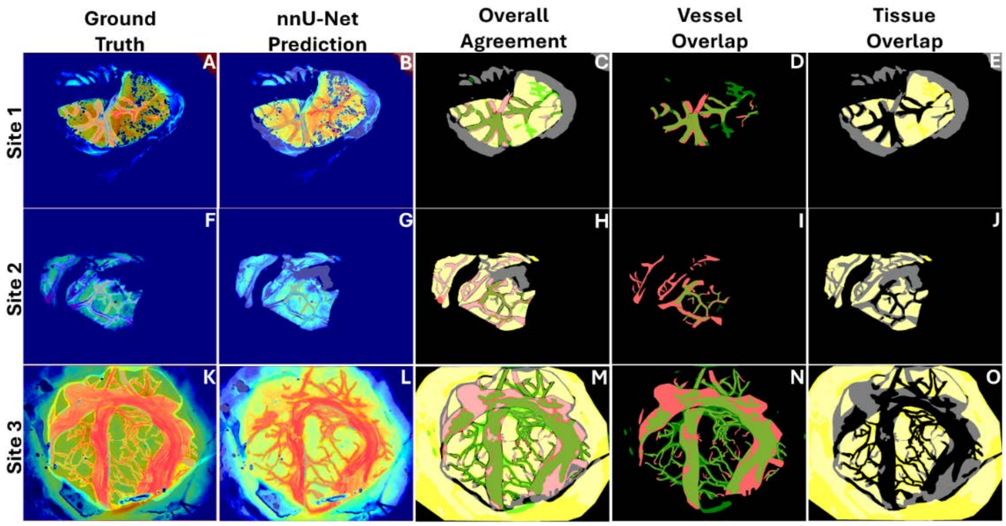

Fig. 6. Qualitative segmentation performance of the nnU-Net model across multi-center intraoperative laser speckle contrast imaging (LSCI) datasets. Representative results are displayed for the Site 1 (A–E), Site 2 (F–J), and Site 3 (K–O) cohorts. The columns correspond to: (A, F, K) ground truth annotations superimposed on LSCI images; (B, G, L) nnU-Net model predictions superimposed on LSCI images; (C, H, M) superposition of model predictions over ground truth labels. In C, H and M, green shows the nnU-Net model predicted blood vessels. Light red shows the ground truth blood vessels. White shows the nnU-Net model predicted cortical tissue. Light yellow shows the ground truth cortical tissues; (D, I, N) segmentation overlap specific to the blood vessel class. In D, I and N, green shows the nnU-Net model predicted blood vessels whereas light red shows the ground truth blood vessels; and (E, J, O) segmentation overlap specific to the cortical tissue class. In E, J and O the white shows the nnU-Net model predicted cortical tissues whereas light yellow shows the ground truth cortical tissues.

across the different site combinations was consistent, with the highest mean validation Dice score observed when training on data from Sites 1 and 3 (0.72), followed by Sites 2 and 3 (0.71), and Sites 1 and 2 (0.68). The relatively small performance variation across these combinations (range: 0.68–0.72) suggests that, although site-specific imaging characteristics modestly affect training dynamics, the model architecture remains stable and effectively integrates data from multiple institutions without reliance on a single dominant site. When applied to the held-out test sets, the performance remained steady. The overall mean test Dice score was 0.67 (0.65 for Site 1, 0.68 for Site 2, and 0.69 for Site 3). Segmentation was strongest for background (0.78–0.95), moderate for cortical tissue (0.66–0.79), and blood vessel performance remained low (0.36–0.49), indicating this class as the main limitation across sites. The consistency between the validation metrics and the test metrics indicates that the model is not overfitting and is capable of handling data from new clinical environments. The slight reduction in performance for the Site 1 testing set (0.65) may be attributed to distinct variations in the surgical field of view or the complexity of the vasculature in those specific cases. Overall, the comparable score across all the institutions confirms that the model is generalizable across the domain shifts and equipment variations inherent in multi-institutional data.

This work aims to bridge the gap between static image analysis and real-time surgical assistance. Delays in image processing can render navigation aids useless during neurosurgical procedures like aneurysm clipping, cerebrovascular bypass, or tumor resection. By establishing that nnU-Net accurately segments static LSCI images, we demonstrate its feature extraction is effective. To

Biomedical Optics EXPRESS

be clinically viable, inference latency must be minimized to ensure the segmented overlay is synchronized with the surgeon's view. The current results demonstrate that the feature extraction capability of the network is sufficient to support architectural optimizations which will reduce computational cost without sacrificing the segmentation quality established in this manuscript.

While the results are promising, there are numerous limitations to this research. The "background" class in this study is highly heterogeneous, containing surgical instruments, gauze, and bone. This variability can occasionally confuse the model, particularly at the boundaries of the surgical field of view. Distinguishing surgical tools from anatomical structures in these boundary regions requires additional spectral features which can be provided by co-registered white-light video. Additionally, LSCI is sensitive to motion artifacts, which were excluded from this static dataset but will be a significant factor in video analysis. Also, specular reflections, which appear as small circular features in LSCI images (blue dots in Fig. 1(B) and black dots in Fig. 1(C)), can degrade the quality of LSCI images and hinder analysis of blood flow changes. Future studies will use a rotatable polarizer during image acquisition to mitigate specular reflections. Further, ground truth labels were generated by a single annotator who is not a neurosurgeon, which introduces the risk of annotation bias and limits assessment of inter-annotator reliability which is a standard concern in clinical segmentation studies.

The dataset of 90 images from 18 unique patients is modest relative to large-scale medical imaging benchmarks, with two main implications. First, it limits the model's exposure to rare intraoperative anatomical and pathological presentations that may be absent from training. Second, it most strongly affects the thin blood-vessel class, where class imbalance and fine structure make performance particularly sensitive to the amount of annotated data; this is reflected in the lower and more variable vessel Dice scores (0.36–0.49) relative to cortical tissue (0.66–0.79) and background (0.78–0.95). To evaluate rather than assume generalizability under these constraints, we adopted a leave-one-site-out scheme in which an entire institution, acquired on a different surgical microscope and from a distinct patient cohort, was withheld for testing in each split. The close agreement between mean validation Dice (0.70) and held-out test Dice (0.67), and the narrow spread across the three held-out sites (0.65–0.69), indicates that the model generalizes across institutions rather than overfitting the small training set. This generalization is further supported by nnU-Net's built-in data augmentation. Prior work shows that clinically useful vessel-segmentation models can be trained on comparably small, annotated datasets; for example, the widely used DRIVE (Digital Retinal Images for Vessel Extraction) retinal vessel dataset contains only 40 images [34]. Nonetheless, larger and more diverse data remain necessary before clinical deployment. Future work will expand the dataset across additional sites and microscope platforms, incorporate multi-annotator labeling to reduce annotation bias and enable inter-annotator reliability assessment, and apply vessel-specific loss weighting to improve the most data-sensitive class.

The smallest detectable vessel diameter is limited by sampling rather than diffraction. At 785 nm through the Kinevo optics, the diffraction-limited spot is on the order of microns, well below our pixel sampling. Instead, the limit is set by the pixel pitch and the spatial-contrast window. With a 3–5 cm field of view across 1280 pixels (the camera used in this study has a pixel resolution of 1280×1024 pixels), each pixel subtends roughly 25–40 μm on the cortex, and the 7×7 window used to compute K yields an effective flow-map resolution element of about 0.2–0.3 mm. A vessel must span at least one resolution element to produce a stable, localized contrast reduction, giving a reliable detection floor of ~0.2 mm; narrower high-flow vessels can still perturb local contrast through partial-volume effects but are not reliably detected. The smallest reliably segmented diameter sits above this detection floor, because a vessel must first be visible in the contrast map, then be annotated, and finally learned by the model; thin vessels near the detection floor are precisely those that are inconsistently labeled and under-detected, which drives the low vessel recall and Dice reported here, placing the reliable segmentation limit somewhat higher; future

Biomedical Optics EXPRESS

studies will aim to quantify this limit. These two limits respond differently to deep learning: the detection limit cannot be lowered by a segmentation network, since no model can recover flow information the sensor never captured, and moves only by changing acquisition parameters (higher optical magnification, finer sampling, or temporal averaging to raise small-vessel signal-to-noise ratio), whereas the segmentation limit can be pushed closer toward the detection floor through vessel-specific and topology-aware loss functions such as centerline Dice, additional thin-vessel training data, multi-annotator labeling, and super-resolution or deconvolution preprocessing.

In conclusion, we demonstrated that the nnU-Net architecture can effectively segment intraoperative LSCI images into distinct anatomical classes. Furthermore, the model effectively handled the inherent noise textures of LSCI images, providing stable segmentation across diverse patient cohorts. This demonstration moves beyond binary blood vessel detection by simultaneously delineating blood vessels, cortical tissue, and background elements. This provides the anatomical context needed to enable automated interpretation of vessel blood flow and cortical tissue perfusion dynamics during microsurgical maneuvers. Future work will focus on expanding the dataset, integrating co-registered white-light video to improve anatomical differentiation, and optimizing inference speed to support low-latency segmentation. Together, these methods will pave the way for neurosurgeons to visualize and quantify blood flow and tissue perfusion by leveraging deep learning to provide real-time video overlays for intraoperative detection of ischemia and hypoperfusion using LSCI, with the potential for empowering safer, more effective operations.

Funding. U.S. National Science Foundation (TRP 26-07).

Acknowledgements. The authors thank the neurosurgical staff at the Barrow Neurological Institute, the Inselspital Bern University Hospital, and the University of Oklahoma College of Medicine. This material is based upon work supported by the U.S. National Science Foundation under the Accelerating Research Translation program 2331409.

Disclosures. Authors A. K. D. and D. R. M. disclose a financial interest in Dynamic Light, Inc. in the form of holding stock, serving on the board of directors, and consulting. No other authors disclose a conflict of interest.

Data availability. Data files analyzed during the current study are available at [35].

Supplemental document. See Supplement 1 for supporting content.

References

1. M. T. Lawton and M. J. Lang, "The future of open vascular neurosurgery: perspectives on cavernous malformations, AVMs, and bypasses for complex aneurysms: JNSPG 75th Anniversary Invited Review Article," J. Neurosurg. **130**(5), 1409–1425 (2019).
2. J. M. Davies and M. T. Lawton, "Advances in open microsurgery for cerebral aneurysms," Neurosurgery **74**(Supplement 1), S7–S16 (2014).
3. E. R. S. Mercyshalinie, A. F. Kharbat, M. Machiorlatti, *et al.*, "Intraoperative tracking of tissue perfusion during cerebral aneurysm surgery with laser speckle contrast imaging: insights beyond standard intraoperative neuromonitoring for detecting ischemia," Neurophotonics **13**(03), 035003 (2026).
4. A. Dimanche, J. Goldberg, D. R. Miller, *et al.*, "Laser speckle contrast imaging versus microvascular Doppler sonography in aneurysm surgery: A prospective study," World Neurosurg X **23**, 100377 (2024).
5. A. Dimanche, D. Bervini, D. R. Miller, *et al.*, "Cortical perfusion measurements with laser speckle contrast imaging during adenosine induced cardiac arrest for aneurysm clipping: a case report," Acta Neurochir (Wien) **166**(1), 27 (2024).
6. A. Dimanche, D. R. Miller, J. Goldberg, *et al.*, "Continuous hemodynamics monitoring during arteriovenous malformation microsurgical resection with laser speckle contrast imaging: case report," Front. Surg. **10**, 1285758 (2023).
7. D. R. Miller, R. Ashour, C. T. Sullender, *et al.*, "Continuous blood flow visualization with laser speckle contrast imaging during neurovascular surgery," Neurophotonics **9**(02), 021908 (2022).
8. J. Goldberg, D. R. Miller, A. Dimanche, *et al.*, "Intraoperative laser speckle contrast imaging to assess vessel flow in neurosurgery: a pilot study," Neurosurgery **94**(5), 983–992 (2024).
9. N. Hecht, M. M. Muller, N. Sandow, *et al.*, "Infarct prediction by intraoperative laser speckle imaging in patients with malignant hemispheric stroke," J Cereb Blood Flow Metab **36**(6), 1022–1032 (2016).
10. N. Hecht, J. Woitzik, J. P. Dreier, *et al.*, "Intraoperative monitoring of cerebral blood flow by laser speckle contrast analysis," Neurosurgical focus **27**(4), E11 (2009).
11. N. Hecht, J. Woitzik, S. König, *et al.*, "Laser speckle imaging allows real-time intraoperative blood flow assessment during neurosurgical procedures," J. Cereb. Blood Flow Metab. **33**(7), 1000–1007 (2013).

Biomedical Optics EXPRESS

12. E. Klijn, H. C. Hulscher, R. K. Balvers, *et al.*, "Laser speckle imaging identification of increases in cortical microcirculatory blood flow induced by motor activity during awake craniotomy," J. Neurosurg. **118**(2), 280–286 (2013).
13. H. van Voorst, J. Su, P. R. Konduri, *et al.*, "Deep generative models for vessel segmentation in CT angiography of the brain," Comput. Biol. Med. **202**, 111432 (2026).
14. N. Passat, C. Ronse, J. Baruthio, *et al.*, "Magnetic resonance angiography: From anatomical knowledge modeling to vessel segmentation," Med. Image Anal. **10**(2), 259–274 (2006).
15. A. El-Baz, A. Elnakib, F. Khalifa, *et al.*, "Precise segmentation of 3-D magnetic resonance angiography," IEEE Trans. Biomed. Eng. **59**(7), 2019–2029 (2012).
16. M. Zhang, C. Zhang, X. Wu, *et al.*, "A neural network approach to segment brain blood vessels in digital subtraction angiography," Computer methods and programs in biomedicine **185**, 105159 (2020).
17. D. Mishra, S. Chaudhury, M. Sarkar, *et al.*, "Ultrasound image segmentation: a deeply supervised network with attention to boundaries," IEEE Trans. Biomed. Eng. **66**(6), 1637–1648 (2019).
18. X. Wang, X. Jiang, and J. Ren, "Blood vessel segmentation from fundus image by a cascade classification framework," Pattern Recognition **88**, 331–341 (2019).
19. S. Lu, C. Y. L. Cheung, J. Liu, *et al.*, "Automated layer segmentation of optical coherence tomography images," IEEE Trans. Biomed. Eng. **57**(10), 2605–2608 (2010).
20. H. Chen, Y. Shi, B. Bo, *et al.*, "Real-time cerebral vessel segmentation in laser speckle contrast image based on unsupervised domain adaptation," Front. Neurosci. **15**, 755198 (2021).
21. E. Morales-Vargas, H. Peregrina-Barreto, R. Q. Fuentes-Aguilar, *et al.*, "Improving blood vessel segmentation and depth estimation in laser speckle images using deep learning," Information **15**(4), 185 (2024).
22. W. Heeman, H. Maassen, K. Dijkstra, *et al.*, "Real-time, multi-spectral motion artefact correction and compensation for laser speckle contrast imaging," Sci. Rep. **12**(1), 21718 (2022).
23. F. Isensee, P. F. Jaeger, S. A. Kohl, *et al.*, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," Nat. Methods **18**(2), 203–211 (2021).
24. G. Dot, T. Schouman, G. Dubois, *et al.*, "Fully automatic segmentation of craniomaxillofacial CT scans for computer-assisted orthognathic surgery planning using the nnU-Net framework," Eur. Radiol. **32**(6), 3639–3648 (2022).
25. E. Von Elm, D. G. Altman, M. Egger, *et al.*, "The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies," The lancet **370**(9596), 1453–1457 (2007).
26. ANSI-Z1361-2007, *American National Standard for the Safe Use of Lasers*, Laser Institute of America, 2007.
27. D. A. Boas and A. K. Dunn, "Laser speckle contrast imaging in biomedical optics," J. Biomed. Opt. **15**(1), 011109 (2010).
28. L. M. Richards, S. S. Kazmi, K. E. Olin, *et al.*, "Intraoperative multi-exposure speckle imaging of cerebral blood flow," J Cereb Blood Flow Metab **37**(9), 3097–3109 (2017).
29. B. C. Russell, A. Torralba, K. P. Murphy, *et al.*, "LabelMe: a database and web-based tool for image annotation," International journal of computer vision **77**(1-3), 157–173 (2008).
30. B. H. Menze, A. Jakab, S. Bauer, *et al.*, "The multimodal brain tumor image segmentation benchmark (BRATS)," IEEE Trans. Med. Imaging **34**(10), 1993–2024 (2015).
31. P. Shi, J. Hu, Y. Yang, *et al.*, "Centerline boundary dice loss for vascular segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2024: Springer, pp. 46–56.
32. A. Tozzi, "Topological analysis of vascular networks: A proof-of-concept study in cerebral angiography," bioRxiv (2025).
33. W. J. Tom, A. Ponticorvo, and A. K. Dunn, "Efficient processing of laser speckle contrast images," IEEE Trans. Med. Imaging **27**(12), 1728–1738 (2008).
34. J. Staal, M. D. Abràmoff, M. Niemeijer, *et al.*, "Ridge-based vessel segmentation in color images of the retina," IEEE Trans. Med. Imaging **23**(4), 501–509 (2004).
35. D. Miller, "Intraoperative LSCI segmentation using deep learning: data," Github, 2026, https://github.com/MillerLabOU/Intraoperative-LSCI-segmentation-using-deep-learning